

Dissociable memory modulation mechanisms facilitate fear amnesia at different timescales

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Abstract

Memory reactivation renders consolidated memory fragile and preludes memory reconsolidation. However, whether memory retrieval facilitates update mechanisms other than memory reconsolidation remains unclear. We tested this hypothesis in three experiments with healthy human participants. First, we demonstrate that memory reactivation prevents the return of fear shortly after extinction training in contrast to the memory reconsolidation effect which takes several hours to emerge and such a short-term amnesia effect is cue independent (Study 1, $N = 57$ adults). Furthermore, memory reactivation also triggers fear memory reconsolidation and produces cue-specific amnesia at a longer and separable timescale (Study 2, $N = 79$ adults). Finally, using continuous theta-burst stimulation (Study 3, $N = 75$ adults), we directly manipulated brain activities in the dorsolateral prefrontal cortex and found that both memory retrieval and intact prefrontal cortex functions were necessary for the short-term fear amnesia. The temporal scale and cue-specificity results of the short-term fear amnesia are clearly dissociable from the amnesia related to memory reconsolidation, and suggest that memory retrieval and extinction training trigger distinct underlying memory update mechanisms. These findings raise the possibility of concerted memory modulation processes related to memory retrieval and shed light to clinical treatment of persistent maladaptive memory.

eLife assessment

The **valuable** findings in this study reveal an intricate pattern of memory expression following retrieval extinction at different intervals from retrieval-extinction to test. The novel advance is in the demonstration that, relative to a standard extinction procedure, the retrieval-extinction procedure more effectively suppresses responses to a conditioned threat stimulus when testing occurs just minutes after extinction. The manuscript provides **incomplete** evidence in support of the attenuation of fear recovery and **solid** evidence for the engagement of the dorsolateral prefrontal cortex in this "short-term" suppression of responding.

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Introduction

Memory retrieval provides a unique window through which consolidated memories can be modified or even neutralized. Studies of memory reconsolidation suggest that memories are rendered fragile each time they are retrieved ^{1,2,3}, possibly due to the unbinding of the old memory trace. The reconsolidation of the new memory trace also requires *de novo* protein synthesis, which usually takes hours to complete. Pharmacologically blocking protein synthesis and behavioral interventions can both eliminate the original fear memory expression in the long-term (24 hours later) memory test retrieval ^{4,5,6,7}, resulting in a cue-specific fear memory deficit ^{2,4,8}. For example, during the reconsolidation window, old fear memory can be updated via extinction training following fear memory retrieval (the retrieval-extinction paradigm) ^{5,6,7}.

However, the exact functional role played by the memory retrieval remains unclear. For example, other than the long-term reconsolidation effects induced by pharmacological manipulations in the fear memory tasks ^{4,9,10}, the short-term amnesia effect of behavioral manipulation such as retrieval-extinction is often overlooked in the fear memory literature ^{5,6,7}. This is in stark contrast with evidence from semantic and episodic memory studies where the behavioral manipulations such as the retrieval-induced forgetting (RIF) protocol or direct suppression would cause short-term memory deficit within 30 minutes ^{11,12}. More recently, behavioral strategies originally developed in the declarative memory studies such as direct suppression had also been successfully applied to the fear memory paradigms in humans and prevented the return of fear shortly afterwards (30mins), suggesting that the fear memory might be amenable to a more immediate effect, in addition to what the memory reconsolidation theory prescribes ¹³.

Parallel to the behavioral manifestation of long- and short-term memory deficits, concurrent neural evidence supporting memory reconsolidation theory emphasizes the long-term effect of memory retrieval by hypothesizing that synapse degradation and *de novo* protein synthesis are required for reconsolidation. These processes typically take several hours, if not longer, and are often aided by overnight sleep ⁹. On the contrary, previous behavioral manipulations engendering the short-term declarative memory effect hinges on the intact activities in brain areas such as dorsolateral prefrontal cortex (cognitive control) and the its functional coupling with specific brain regions such as hippocampus (memory retrieval). The declarative amnesia effect emerges much earlier due to the online functional activity modulation, instead of the much slower protein synthesis in these brain areas ¹³. However, it remains unclear whether memory retrieval might also precipitate a short-term amnesia effect for the fear memory, in addition to the long-term prevention orchestrated by memory reconsolidation.

To fully comprehend the temporal dynamics of the memory retrieval effect, we first carried out a simple fear memory study by pairing a neutral cue (conditioned stimulus, CS) with the electric shock (unconditioned stimulus, US) to directly test whether the fear retrieval-extinction paradigm would yield a short-term effect on fear expression. We then performed another double cue CS–US fear memory experiment to study the temporal characteristics of fear memory malleability to behavioral extinction at the short- (30mins), medium- (6 hours) and long-term (24 hours) time scales.

We hypothesize that the labile state triggered by the memory retrieval may facilitate different memory update mechanisms following extinction training, and these mechanisms can be further disentangled through the lens of temporal dynamics and cue-specificities. Specifically, memory reconsolidation effect will only be evident in the long-term (24h) memory test due to its requirement of new protein synthesis and is cue-dependent¹⁴. In contrast, if more immediate memory update mechanisms are involved, then the fear memory deficit should be observed relatively early (30mins) after extinction training^{15, 16}. There may also be a “limbo” state: when the immediate memory update effects dissipate but reconsolidation updating is not yet fully in effect (6h), the conditioned stimuli should still elicit a fear response, regardless of the reminder (cue). Previous research in the short-term declarative and fear memory deficits highlighted the role of cognitive control. Subjects with better control abilities to the intrusive thoughts or higher functional connectivities between brain regions such as the dorsolateral prefrontal cortex (dlPFC) and hippocampus yielded more severe amnesia in the memory suppression tasks^{13, 17, 35}. Correspondingly, more immediate mechanisms might be associated with individual difference in control abilities over intrusive thoughts¹⁷, which can be measured via the Thought-Control Ability Questionnaire (TCAQ) scale and tested against the individual associative fear memory deficits.

Indeed, through a series of experiments, we identified a short-term fear amnesia effect following memory retrieval, in addition to the fear reconsolidation effect that appeared much later. Finally, by applying the continuous theta-burst stimulation (cTBS) protocol over the dlPFC, the brain region critical for cognitive control and goal maintenance, we showed that both memory retrieval and intact dlPFC function were necessary for the more immediate fear amnesia effect.

Open Practices Statement

Neither of the studies reported in this article was preregistered. The data for both studies are publicly accessible at <https://osf.io/9agvk/>.

Methods

Participants

All participants were students recruited from Peking University. They were right-handed with normal vision and had not participated in electric shock-related experiments before. None of the participants had current or past history of psychiatric illness, nor any contraindication on TMS^{18, 19}. All participants provided informed consent and were remunerated for their participation. This study was approved by the institutional review board of school of psychological and cognitive sciences at Peking University.

We conducted a power analysis (G*Power) to determine the number of participants sufficient to detect a reliable effect²⁰. Based on the effect size of reinstatement effect between treatment group and control group on fear memories reported in the previous literatures ($\eta^2 = 0.19, 0.216, 0.24$; $N = 40, 55, 30$ total sample size, respectively^{9, 13, 21}, 52 participants for both groups (study 1: reminder group and no-reminder group) were enough to detect a significant interaction effect ($\alpha = 0.05, \beta = 0.9, 2$ (groups) $\times 2$ (phases) two-way ANOVA interaction effect). Similarly, a total

of 72 participants for three groups (study 2: 30-min group, 6-h group and 24-h group) were required to detect a reliable effect ($\alpha = 0.05$, $\beta = 0.9$, 3 (groups) $\times$ 2 (phases) $\times$ 2 (CS+) three-way ANOVA interaction effect)²². For the rTMS study, based on the effect size of reinstatement effect between treatment group and control group on fear memories reported in the previous literatures ($\eta^2 = 0.144, 0.19, 0.216, 0.24$; $N = 79, 40, 55, 30$ total sample size, respectively^{9, 13, 21}), a total of 72 participants (study 3) were needed to detect a significant interaction effect ($\alpha = 0.05$, $\beta = 0.8$, 4 (groups) $\times$ 2 (phases) $\times$ 2 (CS+) three-way ANOVA interaction effect).

In study 1, we first recruited a total of 75 human subjects (41 females; mean age = 21.3, SD = 2.21). Eight participants discontinued after Day 1 due to their non-measurable spontaneous conditioned stimulus (CS) signal (the skin conductance response, SCR) towards different CS (non-responders, SCR response to any CS was less than 0.02 μ S, $n = 0$ & 8; reminder group and no-reminder group, respectively). Ten participants (7 females; mean age = 20.6, SD = 1.79) who were “non-learners” during fear acquisition ($n = 3$ & 1 in both groups; respectively) or extinction ($n = 3$ & 3, respectively) phase were also excluded from further analysis. The criteria for “non-learners” in the fear acquisition include smaller mean SCR of CS+ (CS that was partially paired with the electric shock) than that of the CS- (CS that was never paired with shock) in the latter half trials of acquisition, and a smaller mean SCR difference between CS+ and CS- in the latter than the first half trials of fear acquisition on day 1. Similarly, the criteria for “non-learners” in the fear extinction are larger mean SCRs of CS+ than the CS- responses in both the last trial and the latter half trials of extinction and a larger mean SCR difference between CS+ and CS- in the latter than the first half trials during fear extinction on day 2 (see the exclusion criteria Table S1 in the Supplemental Material). Therefore, we had a total of 57 subjects for data analysis in study 1 (30 for the reminder group, 17 females, mean age = 21.4, SD = 2.37, and 27 for the no-reminder group, 15 females, mean age = 21.62, SD = 2.13). The group $\times$ gender interaction effect on participants’ age was not statistically significant ($F_{1,49} = 0.137$, $P = 0.713$, $\eta^2 = 0.003$, two-way ANOVA), and gender ($\chi^2 = 0.007$, $P = 0.933$) and age ($t_{55} = -0.386$, $P = 0.708$) were not significantly different between two groups.

In study 2, we first recruited 119 participants (56 females; mean age = 20.85, SD = 2.59). Thirty-seven participants discontinued after Day 1 for they were “non-responders” ($n = 19, 5$ and 13; 30-min group, 6-h group and 24-h group, respectively). Additionally, two participants ($n = 1$ & 1; 30-min group and 24-h group; respectively) failed to show the evidence of fear acquisition and one participant (30-min group) failed to show the evidence of fear extinction on day 2 and were also excluded from further analysis. Finally, we had a total of 79 subjects for data analysis in study 2 (27 for the 30-min group, 15 females, mean age = 21.4, SD = 2.46; 26 for the 6-h group, 12 females, mean age = 20.54, SD = 1.74; and 26 for the 24-h group, 12 females, mean age = 20.96, SD = 2.990). The interaction effect of gender $\times$ group on participants’ age was not statistically significant ($F_{2,73} = 0.213$, $P = 0.809$, $\eta^2 = 0.006$, two-way ANOVA), and gender ($\chi^2 = 0.021$, $P = 0.990$) or age ($F_{2,76} = 0.661$, $P = 0.519$, $\eta^2 = 0.017$) was not significantly different across three groups.

In study 3, we recruited a total of 90 participants (47 females; mean age = 21.01, SD = 2.44). Fifteen participants (8 females; mean age = 21.2, SD = 1.22) were excluded from further analysis since they failed to show SCR response to the any stimuli (CS1+, CS2+ or CS-) (non-responders, $n = 3, 4, 4, 4$; R rdLPFC group, R vertex group, NR rdLPFC group and NR vertex group, respectively, see Supplemental Table S1)^{6, 7, 23–28}. Our final sample included 75 participants for data analysis (19 for the Reminder right dLPFC group, 10 females, mean age = 22.05, SD = 2.44; 18 for the Reminder vertex group, 9 females, mean age = 20.67, SD = 2.68; 18 for the No-reminder right dLPFC group, 10 females, mean age = 20.28, SD = 2.85; and 20 for the No-reminder vertex group, 10 females, mean age = 20.85, SD = 2.43). A two-way ANOVA showed no significant effect of age of TMS ($F_{1,71} = 0.459$, $P = 0.500$, $\eta^2 = 0.006$), reminder ($F_{1,71} = 1.756$, $P = 0.189$, $\eta^2 = 0.024$), nor their interaction ($F_{1,71} = 2.659$, $P = 0.107$, $\eta^2 = 0.036$).

Conditioned stimuli and psychophysiological stimulation

For study 1, we employed two squares with different colors (yellow and blue) that served as CS+ and CS-. During fear acquisition, the CS+ was paired with a mild electrical shock (US) on a 37.5% partial reinforcement scheme (6 CS+ paired with electric shock and 10 CS+ with no shock), and the CS- was never paired with a shock (10 CS-). A pseudo-random CS delivery order was generated for the fear acquisition, extinction and test phase for two groups with the rule that no same trial-type (CS+ and CS-) repeated more than twice. Assignment of square colors to the CS (CS+ and CS-) was counterbalanced across participants.

For study 2 & 3, three squares with different colors (yellow, red and blue) were used as CS1+, CS2+ and CS-, respectively. During the fear acquisition phase, both CS1+ and CS2+ were paired with electric shocks (US) on a 37.5% partial reinforcement scheme, and the CS- was not paired with shocks (10 non-reinforced presentations of CS1+, CS2+ and CS- each, intermixed with an extra of 6 CS1+ and 6 CS2+ that co-terminated with the shock). Again, a pseudo-random stimulus order was generated for fear acquisition and extinction phases of three groups with the rule that no same trial-type (CS1+, CS2+ and CS-) repeated more than twice. In the test phase, to exclude the possibility that the difference between CS1+ and CS2+ was simply caused by the presentation sequence of CS1+ and CS2+, half of the participants completed the test phase using a pseudo-random stimuli sequence and the identities of CS1+ and CS2+ reversed in the other half of the participants. For all three groups, the CS colors were counterbalanced across participants.

The US was a mild electrical shock delivered to the right wrist of participants via a DS-5 Isolated Bipolar Constant Current Stimulator (Digitimer, Welwyn Garden City, UK). The US levels were set by the participants themselves, starting from a low level of shock (5v) and gradually increased and settled to a level that they described as ‘uncomfortable, but not painful’ (with the maximum level of 10v). The duration of all electric shocks was 200ms with 50 pulses per second.

SCR measurements were collected using two Ag-AgCl electrodes attached to the tips of the index and middle fingers of each subject’s left hand. All skin conductance data were recorded via the Biopac[®] MP160 BioNomadix System, and analyzed using the Acknowledgement 5.0 software. The SCR level for each trial was calculated as the amplitude difference (in microsiemens) from peak to trough during the 0.5 to 4.5s window after the CS (CS SCR) or US (US SCR) onset and responses below 0.02 μ S were encoded as zero. The raw SCR scores were divided by each subject’s mean US responses and then square-root-transformed to normalize SCR across individuals and groups (Wang et al., 2021).

Continuous theta burst stimulation (cTBS) was applied with a Magstim Rapid 2 system (Magstim, Whitland, Wales, UK). Before the experiments, we determined the stimulation intensity of the cTBS protocol that was at 80% of the individual active motor threshold (AMT). Each transcranial magnetic stimulation (TBS) burst consisted of three stimuli pulses at 50HZ, with each train being repeated every 200ms (5Hz). In this cTBS protocol, a 40s train of uninterrupted TBS is given (600 pluses)²⁴²⁵. During the stimulation, the cTBS was applied either over the right dlPFC or the vertex, which was determined by standard F4 and Cz locations based on international electroencephalogram 10-20 system measurements²⁹.

The control ability over intrusive thoughts was measured by the 25-item Thought-Control Ability Questionnaire (TCAQ) scale³⁰. Participants were asked to rate on a five-point Likert-type scale the extent to which they agreed with the statement from 1 (*completely disagree*) to 5 (*completely agree*). At the end of the experiments, all participants completed the TCAQ scale to assess their perceived control abilities over intrusive thoughts in daily life¹⁷. The internal consistency (Cronbach’s alpha) of TCAQ scale was 0.92, and the reliability coefficient was satisfactory ($r = 0.88$).

Experimental procedure

Study 1, 2 and 3 were carried out for two or three consecutive days with three phases. Before the experiments, all participants gave informed consent. During the experiments, subjects were required to stay relaxed and still, focus on the computer screen and pay attention to the relationship between color stimuli (CS) and the electric shock (US).

Day 1: Acquisition. For study 1, two groups of participants underwent a fear-conditioning paradigm (acquisition) where the CS+ was partially (37.5%) paired with the US and the CS- was never paired with the shock. For study 2 & 3, participants underwent fear conditioning using three color squares as the CS (CS1+, CS2+ and CS-). The CS were presented for 4s each with a 10-12s variable ITI in all the studies (1, 2 and 3).

Day 2: Retrieval-extinction phase. For study 1, the fear memory was reactivated before extinction in the reminder group. During reactivation, the CS+ was presented once without the delivery of the US. Followed by a 10-min break, the reminder group underwent extinction training with 10 CS+ and 11 CS- presentation non-reinforced. In the no-reminder group, the fear memory was not reactivated (no CS+ reminder). To match with the timeline of the reminder group, the no-reminder group subjects still needed to wait for 10-min after the beginning of the experiment on day 2 and then underwent extinction training with 11 CS+ and 11 CS+ non-reinforced presentation. During the 10-min break, all participants were asked to rest and stay still.

For study 2, only one of the CS+ (CS1+) was presented once without the US during reactivation in all three groups. After a 10-min break, all three groups underwent extinction training with 10 CS1+, 11CS2+ and 11CS- non-reinforced presentation.

For study 3, for the reminder groups (reminder-dlPFC and reminder-vertex groups), only one of the CS+ (CS1+) was presented once (without the US delivery) as the retrieval trial. No CS+ reminder was delivered for the no-reminder groups (no-reminder dlPFC and no-reminder vertex groups). Additionally, during the 10-min break (8 mins after the supposed CS+ reminder delivery), cTBS was applied either over the right dlPFC (dlPFC groups: PFC) or the vertex (vertex groups: VER), resulting in a 2 (reminder vs. no-reminder) x 2 (dlPFC vs. vertex) of 4 different groups (reminder-dlPFC: R-PFC, reminder-vertex: R-VER, no-reminder dlPFC: NR-PFC, no-reminder vertex: NR-VER).

Day 2 or Day 3: Reinstatement and test phases. For study 1, 30 minutes after the extinction training, both groups received four un-signalized electric shocks with 10s to 12s ITI (reinstatement). Eighteen seconds after the reinstatement phase, all subjects were presented with CS+ and CS- 10 times each without electric shocks and their SCRs were recorded (test phase).

For study 2, participants were assigned into three groups: the 30min group, the 6h group and the 24h group. All participants received four un-signalized electric shocks with 10s to 12s ITI (reinstatement) after their corresponding time delays (30min, 6h or 24h) between extinction and reinstatement. Eighteen seconds after the reinstatement phase, all subjects were tested via the presentation of CS1+, CS2+ and CS- 10 times each without electric shocks associated.

For study 3, the reinstatement and testing procedures were the same as in study 2 except that they were conducted one hour after the extinction training.

Results

The short-term effect of the fear memory retrieval-extinction procedure

To test the memory-retrieval-induced fear memory deficits at different time scales in humans, we designed two experiments to examine whether the fear memory retrieval-extinction procedure would block the return of fear memory.

In the first study, we aimed to test whether there is a short-term amnesia effect of fear memory retrieval following the fear retrieval-extinction paradigm. We measured subjects' SCRs for the fear acquisition, extinction and test phases to assess the associative fear memory and the recovery of fear memory. At each phase, the differential fear response was calculated as the difference between SCRs to the CS+ and CS-.

To assess fear acquisition across groups (**Fig. 1b&c**), we conducted a mixed two-way ANOVA (reminder vs. no-reminder) x time (early vs. late part of the acquisition) on the differential fear SCR. Our results showed a significant main effect of time (early vs. late; $F_{1,55} = 6.545$, $P = 0.013$, $\eta^2 = 0.106$), suggesting successful fear acquisition in both groups. There was no main effect of group (reminder vs. no-reminder) or the group x time interaction (group: $F_{1,55} = 0.057$, $P = 0.813$, $\eta^2 = 0.001$; interaction: $F_{1,55} = 0.066$, $P = 0.798$, $\eta^2 = 0.001$), indicating similar levels of fear acquisition between two groups. Post-hoc *t*-tests confirmed that the fear responses to the CS+ were significantly higher than that of CS- during the late part of acquisition phase in both groups (reminder group: $t_{29} = 6.642$, $P < 0.001$; no-reminder group: $t_{26} = 8.522$, $P < 0.001$; **Fig. 1c**). Importantly, the levels of acquisition were equivalent in both groups (early acquisition: $t_{55} = -0.063$, $P = 0.950$; late acquisition: $t_{55} = -0.318$, $P = 0.751$; **Fig. 1c**).

To ensure equivalent and successful extinction across groups (**Fig. 1b & d**), we performed similar mixed two-way ANOVA (group x trial) on the differential SCR in the extinction phase. Our results showed significant main effect of trial ($F_{10,550} = 2.825$, $P = 0.002$, $\eta^2 = 0.049$), but no effect of group ($F_{1,55} = 0.063$, $P = 0.802$, $\eta^2 = 0.001$) or their interaction ($F_{10,550} = 1.77$, $P = 0.063$, $\eta^2 = 0.031$). Follow-up *t*-tests further confirmed that in the last trial of extinction there was no difference between fear responses to CS+ and CS- within group (reminder group: $t_{29} = -1.147$, $P = 0.261$; no-reminder group: $t_{26} = 0.261$, $P = 0.796$; **Fig. 1d**). Importantly, the levels of extinction did not show significant difference between two groups ($t_{55} = -1.074$, $P = 0.288$; **Fig. 1d**).

Fear recovery was assessed using a two-way ANOVA with main effects of group (reminder vs. no-reminder) and time (last trial of extinction vs. first trial of test) and it showed a significant time x group interaction ($F_{1,55} = 4.087$, $P = 0.048$, $\eta^2 = 0.069$). We defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS. Post hoc *t*-tests showed that fear memories were resilient after regular extinction training, as demonstrated by the significant difference between fear recovery indexes of the CS+ and CS- for the no-reminder group ($t_{26} = 7.441$, $P < 0.001$; **Fig. 1e**), while subjects in the reminder group showed no difference of fear recovery between CS+ and CS- ($t_{29} = 0.797$, $P = 0.432$, **Fig. 1e**). Finally, there is no statistical difference between the differential fear recovery indexes between CS+ in the reminder and no-reminder groups ($t_{55} = -2.022$, $P = 0.048$; **Fig. 1c**, also see Supplemental Material for direct test for the test phase). Also, our results remain robust even with the “non-learners” included in the analysis (Fig. S1 in the Supplemental Material).

Together, these results indicate that the retrieval before extinction procedure prevents the return of fear expression in the short-term test, distinct from the well-established fear reconsolidation effect at a much later time ⁷. In the current study, fear recovery was tested 30 minutes after

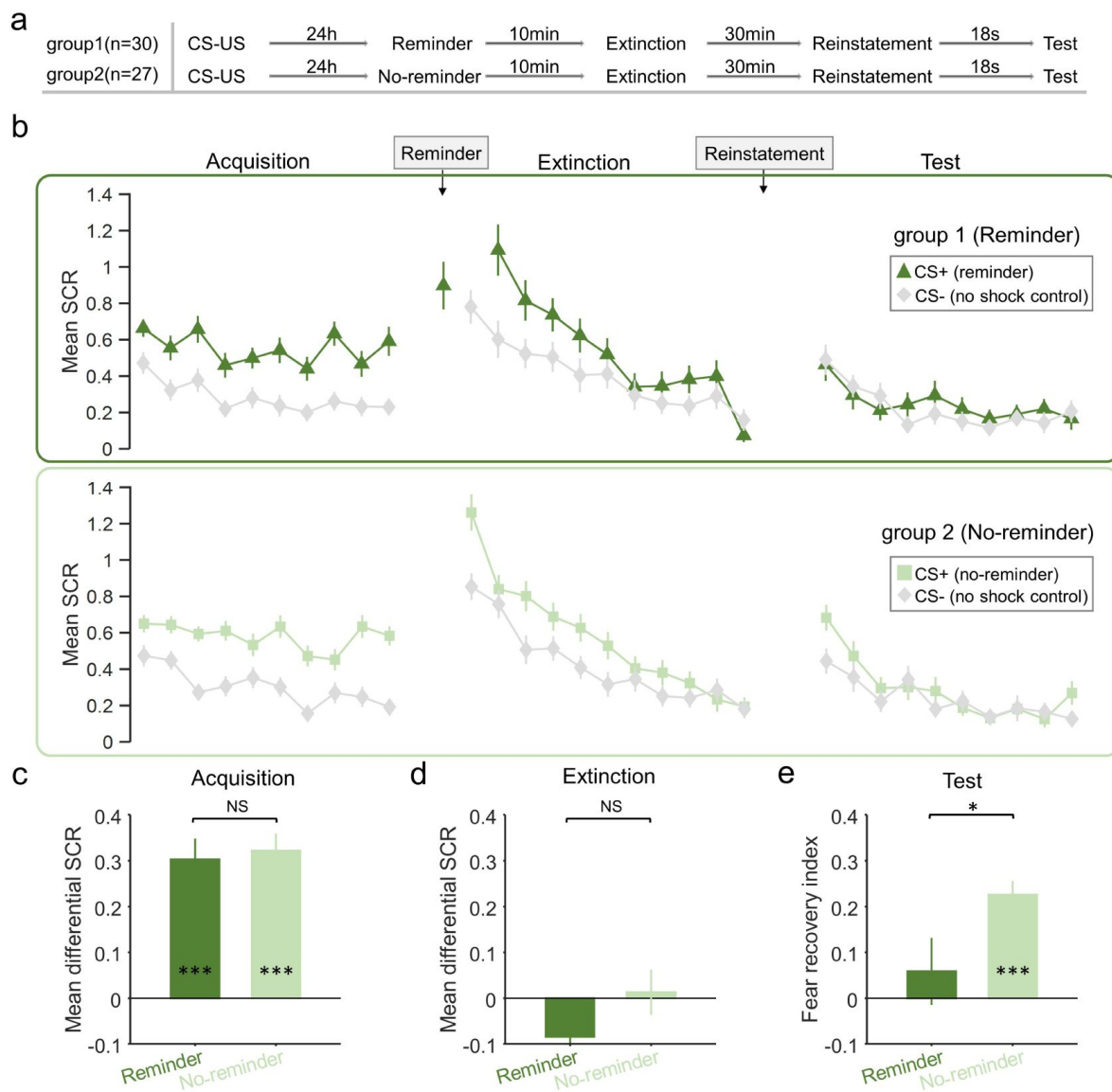


Fig. 1 |

SCR responses to conditioned stimuli in two groups of study 1.

a, Experimental design and timeline. **b**, Mean SCRs of fear conditioned stimuli (CS+) and the control stimulus (CS-) across fear acquisition, extinction and test phases for two groups (reminder and no-reminder). **c**, Mean differential SCRs (CS+ minus CS-) in the acquisition phase (latter half trials). **d**, Mean differential SCRs (CS+ minus CS-) in the extinction phase (last trial). **e**, Differential fear recovery index between CS+ and CS- in the test phase. *** $P < 0.001$, * $P < 0.05$. NS: Non-significant. Error bars represent standard errors.

extinction training, whereas the effect of memory reconsolidation was generally evident only several hours later and possibly with the help of sleep ^{9,10}, leaving open the possibility of a different cognitive mechanism for the short-term fear dementia related to the retrieval-extinction procedure. Importantly, such a short-term effect is also retrieval dependent, suggesting the labile state of memory is necessary for the short-term memory update to take effect (**Fig. 1e**).

Temporal and generalizing characteristics of the retrieval-extinction effect

Given the findings of short-term fear amnesia following the retrieval-extinction paradigm, we set out to map the temporal dynamics of fear amnesia as well as its cue-specificities - whether the amnesia triggered by a specific CS+ reminder (for example CS1+) generalizes to the fear memory associated with the other CS+ (CS2+). We adopted a double-cue fear learning paradigm and examined fear recovery in the short- (30 mins), medium- (6 hours) and long-term (24 hours) fear memory test in three separate groups of participants after the extinction training (**Fig. 2a**).

Fear acquisition was assessed using a mixed three-way ANOVA with group (30min, 6h and 24h), CS+ (CS1+ vs. CS2+, defined as the mean SCR differences between each CS+ and CS-) and trial numbers (10 trials) as main factors. This analysis showed a significant main effect of trial ($F_{9,684} = 12.782, P < 0.001, \eta^2 = 0.144$), but no effect of group ($F_{2,76} = 2.121, P = 0.127, \eta^2 = 0.053$), CS+ ($F_{1,76} = 0.575, P = 0.45, \eta^2 = 0.008$) or their interactions (all P s > 0.05; **Fig. 2b**). Furthermore, to ensure equivalent fear acquisition across three groups, a mixed two-way ANOVA showed no effect of group (30min, 6h and 24h; $F_{2,76} = 1.79, P = 0.174, \eta^2 = 0.045$), CS+ (CS1+ vs. CS2+; $F_{1,76} = 1.337, P = 0.251, \eta^2 = 0.017$) or their interaction ($F_{2,76} = 0.959, P = 0.388, \eta^2 = 0.025$) on the late phase of acquisition (last 5 trials). Post-hoc t -tests confirmed that the SCRs elicited by CS1+ and CS2+ were significantly higher than that of CS- among three groups on the late phase of acquisition (CS1+: $t_{26} = 6.806, P < 0.001$; CS2+: $t_{26} = 6.853, P < 0.001$ for the 30min group; CS1+: $t_{25} = 9.822, P < 0.001$; CS2+: $t_{25} = 9.841, P < 0.001$ for the 6h group and CS1+: $t_{25} = 6.909, P < 0.001$; CS2+: $t_{25} = 6.956, P < 0.001$ for the 24h group), indicating successful and similar fear acquisition across all groups (**Fig. 2c**).

We then set out to examine whether participants underwent successful extinction training on day 2. The similar mixed three-way ANOVA showed a significant effect of trial ($F_{10,760} = 8.219, P < 0.001, \eta^2 = 0.098$), but no effect of group ($F_{2,76} = 1.822, P = 0.169, \eta^2 = 0.046$), CS+ ($F_{1,76} = 0.91, P = 0.343, \eta^2 = 0.012$) or their interactions (all P s > 0.1; **Fig. 2b**). To test whether participants across all groups achieved similar levels of extinction, a mixed two-way ANOVA showed no effect of group (30min, 6h and 24h; $F_{2,76} = 0.059, P = 0.943, \eta^2 = 0.002$), CS+ (CS1+ vs. CS2+; $F_{1,76} = 0.258, P = 0.613, \eta^2 = 0.003$) or their interaction ($F_{2,76} = 0.504, P = 0.606, \eta^2 = 0.013$) on the last trial of extinction training. Post-hoc t -tests showed that there was no difference between CS+ and CS- responses in all groups on the last trial of extinction (CS1+: $t_{26} = -0.397, P = 0.695$; CS2+: $t_{26} = -0.505, P = 0.618$ for the 30min group; CS1+: $t_{25} = -0.054, P = 0.957$; CS2+: $t_{25} = 0.140, P = 0.890$ for the 6h group and CS1+: $t_{25} = 0.435, P = 0.667$; CS2+: $t_{25} = -0.317, P = 0.754$ for the 24h group; **Fig. 2d**).

To assess the effects of the retrieval-extinction procedure as a function of the time delay between fear extinction and test phases, we conducted a 3-way ANOVA with the within-subject factor CS+ (CS1+ vs. CS2+, defined by the mean SCR differences between each CS+ and CS-), time (last trial of extinction vs. first trial of test) and between-subject factor group (30min, 6h and 24h) and found a significant three-way interaction ($F_{2,76} = 6.376, P = 0.003, \eta^2 = 0.144$). To disentangle this interaction, we further examined the effects in each group.

In the 30min group, there was no significant effect of time ($F_{1,26} = 0.208, P = 0.652, \eta^2 = 0.008$), CS+ ($F_{1,26} = 0.888, P = 0.355, \eta^2 = 0.033$) or their interaction ($F_{1,26} = 1.039, P = 0.317, \eta^2 = 0.038$). Post-hoc t -tests showed that the retrieval-extinction training diminished fear responses to both the reminded CS+ (differential fear recovery index between CS1+ and CS-, $t_{26} = -0.787, P = 0.438$; **Fig.**

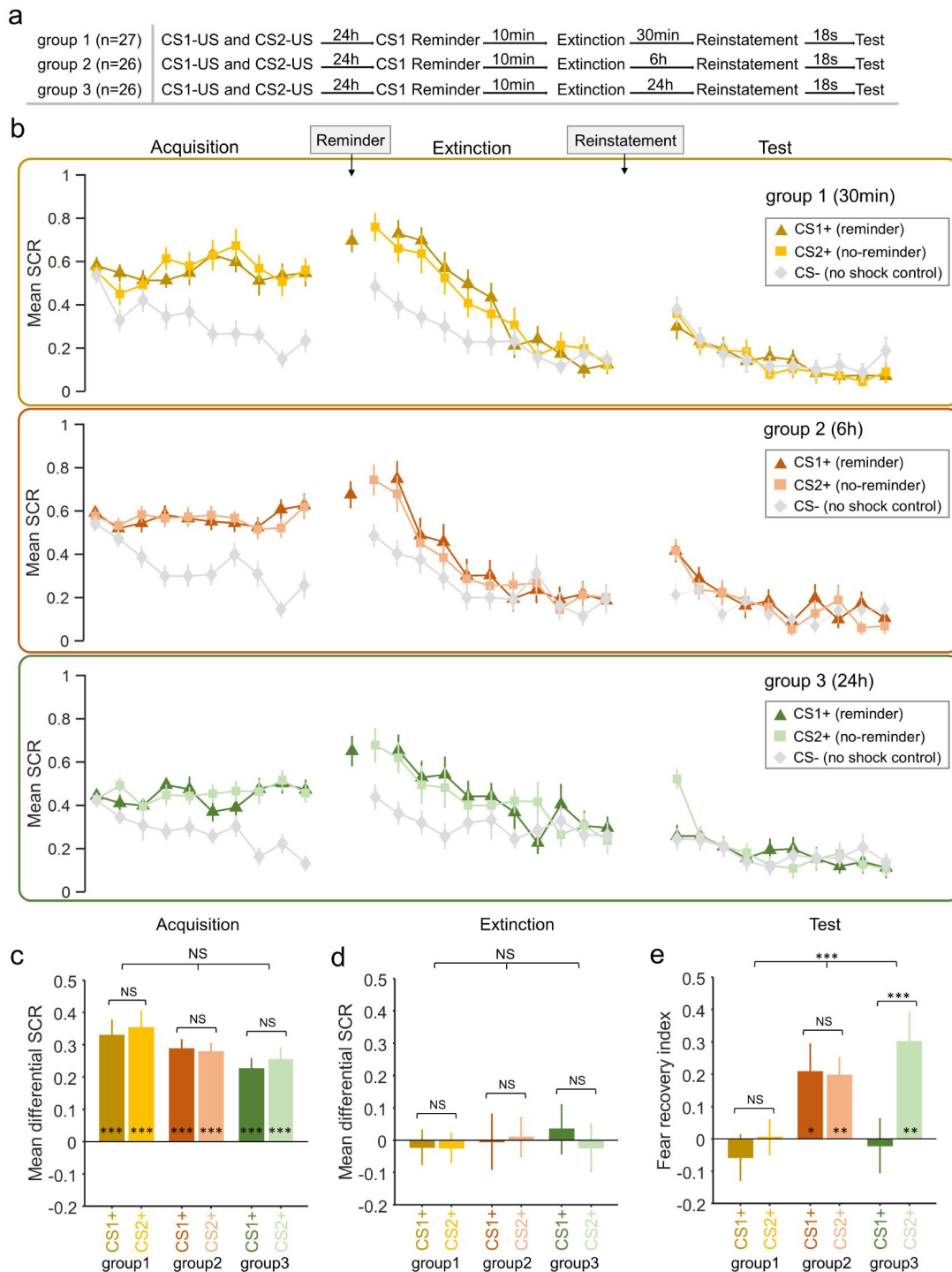


Fig. 2 |

SCR responses to conditioned stimuli in three groups of study 2.

a, Experimental design and timeline. **b**, Mean SCRs of fear conditioned stimuli CS1+ (reminder) and CS2+ (No-reminder), and the control stimulus (CS-) across the fear acquisition, extinction and test phases for each group (30min, 6h and 24h tests). **c**, Mean differential SCRs (CS1+ minus CS- and CS2+ minus CS-) in the acquisition phase (latter half trials). **d**, Mean differential SCRs in the acquisition phase (latter half trials). **e**, Mean differential SCRs in the extinction phase (last trial). **f**, Differential fear recovery index between CS+ and CS- in the test phase. *** $P < 0.001$. * $P < 0.05$. NS: Non-significant. Error bars represent standard errors.

2c) and the non-reminded CS+ (differential fear recovery index between CS2+ and CS-, $t_{26} = 0.088$, $P = 0.93$; **Fig. 2e**), suggesting a cue-independent short-term amnesia effect. This result indicates that the short-term amnesia effect observed in Study 2 is not reminder-cue specific and can generalize to the non-reminded cues. In addition, there was no significant difference between the fear recovery index associated with CS1+ and CS2+ ($t_{26} = -1.019$, $P = 0.317$; **Fig. 2e**, also see the supplemental material for direct test of the test phase SCRs).

In contrast to the short-term effect, there was a significant effect of time ($F_{1,25} = 9.654$, $P = 0.005$, $\eta^2 = 0.279$), but no effect of CS+ ($F_{1,25} = 0.036$, $P = 0.85$, $\eta^2 = 0.001$) or their interaction ($F_{1,25} = 0.028$, $P = 0.869$, $\eta^2 = 0.001$) in the 6h (medium-term) group. Post-hoc *t*-tests showed that fear memory recovered for the reminded CS+ (differential fear recovery index between CS1+ and CS-, $t_{25} = 2.382$, $P = 0.05$), as well as the non-reminded CS+ (differential fear recovery index between CS2+ and CS-, $t_{25} = 3.525$, $P = 0.004$) in the medium-term test of the retrieval-extinction procedure effect, suggesting the failure to suppress the return of extinguished fear memory at such a time scale (**Fig. 2e**). There was also no significant difference measured by the fear recovery index between CS1+ and CS2+ ($t_{25} = 0.166$, $P = 0.869$; **Fig. 2e**).

Finally, in line with previous research on fear memory reconsolidation, significant SCR difference of the fear reinstatement effect between the reminded CS+ and non-reminded CS+ (time $\times$ CS+ interaction: $F_{1,25} = 16.924$, $P < 0.001$, $\eta^2 = 0.404$) was detected in the 24h group via two-way ANOVA. More specifically, the retrieval-extinction procedure only diminished fear responses to the reminded CS+ (differential fear recovery index between CS1+ and CS-, $t_{25} = 0.25$, $P = 0.805$), whereas the fear response to the non-reminded CS+ remained significant (differential fear recovery index between CS2+ and CS-, $t_{25} = 3.269$, $P = 0.009$; **Fig. 2e**). Post-hoc *t*-test showed that the fear recovery index between CS1+ and CS2+ was significantly different ($t_{25} = -4.114$, $P = 0.001$; **Fig. 2c**), aligning well with previous literature that suggested the cue-specificity of the fear reconsolidation effect ^{5,14}. Similarly, these results remain robust even with the “non-learners” included in the analysis (Fig. S2 in the Supplemental Material).

Altogether, in line with our hypothesis, the return of fear memory measured by the reinstatement effect showed distinct patterns at different time scales. The retrieval-extinction procedure diminished fear SCRs for both the reminded and non-reminded CS+, engendering cue-independent amnesia in the short-term test (30min). On the other hand, at the long-term (24h) fear memory test, fear responses were only eradicated for the reminded CS+ (but not the non-reminded CS+) with a cue-specific effect, consistent with previous literatures on memory reconsolidation. Interestingly, SCR measures of the fear memory re-appeared in the medium-term (6h) test and there was no difference between fear responses elicited by the reminded and non-reminded CS+, suggesting that the 6-hour time-scale may be a “limbo” time point where both the immediate memory update mechanism and memory reconsolidation are not effective.

Fear amnesia effect as a function of the thought-control ability

Finally, to assess whether the fear reinstatement effects at different time scale is related to the thought-control ability, we first ran a one-way ANOVA to confirm that subjects among different groups did not differ in their thought-control abilities according to the TCAQ scores ($F_{2,76} = 0.154$, $P = 0.857$, $\eta^2 = 0.004$). We then ran separate linear regressions between the differential fear recovery index (differential fear recovery index between CS+ and CS-) and thought-control abilities (TCAQ scores) in the three groups. This analysis yielded significant correlation between TCAQ score and fear recovery index for both CS+ in the 30min group (**Fig. 3a**, main effect of thought-control ability, $t_{51} = -3.446$, $P = 0.003$, Bonferroni correction), indicating that subjects endowed with higher thought-control abilities were less likely to experience the return of fear response. However, no significant correlation was observed in the 6h or the 24h group (**Fig. 3b & c**; P s > 0.4). Post-hoc analysis of the 30min group showed significant correlation between the fear recovery index of

both CS1+ and CS2+ with subjects' thought-control abilities (**Fig. 3a**, CS1+: $r = -0.462$, $P = 0.015$; CS2+: $r = -0.413$, $P = 0.032$). No significant TCAQ and CS+ interaction effect in the three groups (P s > 0.4).

It is worth noting that the correlations between TCAQ scores and fear responses in all three groups were not significant in the late phase of acquisition or the last trial of extinction (all P s > 0.10), indicating that the effect of thought-control ability was specifically reflected in the fear recovery index.

In conclusion, thought-control ability only affected fear recovery in the short-term (30min) test for both the reminded and non-reminded CS+; however, the association between thought-control ability and fear recovery index was not identified at longer time scales (6h and 24h). These results are consistent with previous research which suggested that people with better capability to resist intrusive thoughts also performed better in motivated dementia in both declarative and associative memories ^{13, 17}.

dIPFC dependent short-term fear memory amnesia

In study 3 (**Fig. 4a**), we tested that in addition to the memory retrieval cue (reminder), whether the intact right dIPFC activity was necessary for the short-term fear amnesia. To test this hypothesis, we set up a reminder CS+ (CS1+ – CS– vs. CS2+ – CS–) x reminder (reminder vs. no-reminder) x cTBS (rdIPFC vs. vertex) x trial number four-way ANOVA against the SCRs for the acquisition and extinction phases. In the acquisition phase, as expected, there were no significant main effects of reminder ($F_{1,71} = 1.291$, $P = 0.260$, $\eta^2 = 0.018$), cTBS ($F_{1,71} < 0.001$, $P = 0.990$, $\eta^2 < 0.001$), CS+ ($F_{1,71} = 1.927$, $P = 0.169$, $\eta^2 = 0.026$) or their interactions (all P s > 0.1; **Fig. 4b&c**) but a significant main effect of trial number ($F_{9,639} = 8.054$, $P < 0.001$, $\eta^2 = 0.102$), indicating successful learning of CS+ stimuli. Indeed, post-hoc tests confirmed that the SCRs of CS+ was significantly higher than that of the CS– on the late phase (last 5 trials) of acquisition in all four groups (CS1+: $t_{18} = 7.735$, $P < 0.001$; CS2+: $t_{18} = 8.546$, $P < 0.001$ for the R-PFC group; CS1+: $t_{17} = 8.568$, $P < 0.001$; CS2+: $t_{17} = 8.478$, $P < 0.001$ for the R-VER group; CS1+: $t_{17} = 6.032$, $P < 0.001$; CS2+: $t_{17} = 7.722$, $P < 0.001$ for the NR-PFC group and CS1+: $t_{19} = 9.110$, $P < 0.001$; CS2+: $t_{19} = 6.538$, $P < 0.001$ for the NR-VER group, **Fig. 4c**).

The effects of cTBS over the right dIPFC after the memory reactivation were assessed using the similar mixed-effect four-way ANOVA, which showed significant effects of trial ($F_{10,710} = 8.566$, $P < 0.001$, $\eta^2 = 0.108$), cTBS ($F_{1,71} = 6.269$, $P = 0.015$, $\eta^2 = 0.081$) and cTBS x trial interaction ($F_{10,710} = 2.292$, $P = 0.012$, $\eta^2 = 0.031$), but no effect of reminder ($F_{1,71} = 1.894$, $P = 0.173$, $\eta^2 = 0.026$), CS+ ($F_{1,71} = 0.398$, $P = 0.530$, $\eta^2 = 0.006$) or other interactions (all P s > 0.1; **Fig. 4b**). To test whether participants across all groups achieved similar levels of extinction, a mixed-effect three-way ANOVA was performed on the last trial of extinction training and there was no significant effect of reminder (reminder and no-reminder; $F_{1,71} = 3.112$, $P = 0.082$, $\eta^2 = 0.043$), cTBS ($F_{1,71} = 0.929$, $P = 0.338$, $\eta^2 = 0.013$), CS+ (CS1+ vs. CS2+; $F_{1,71} = 1.623$, $P = 0.207$, $\eta^2 = 0.022$) or their interactions (all P s > 0.1; **Fig. 4d**). Furthermore, post-hoc t -tests showed that there was no difference between CS+ and CS– responses in all groups on the last trial of extinction (CS1+: $t_{18} = 1.310$, $P = 0.207$; CS2+: $t_{18} = 1.297$, $P = 0.211$ for the R-PFC group; CS1+: $t_{17} = 0.839$, $P = 0.413$; CS2+: $t_{17} = 1.090$, $P = 0.291$ for the R-VER group; CS1+: $t_{17} = 0.380$, $P = 0.709$; CS2+: $t_{17} = 0.624$, $P = 0.541$ for the NR-PFC group and CS1+: $t_{19} = -1.434$, $P = 0.168$; CS2+: $t_{19} = -0.535$, $P = 0.599$ for the NR-VER group; **Fig. 4d**).

To assess the effects of the right dIPFC on the memory retrieval related short-term amnesia, we conducted a mixed-effect 3-way ANOVA with the within-subject factors CS+ (CS1+ vs. the CS2+), between-subject factors reminder (reminder vs. no-reminder) and cTBS (dIPFC vs. vertex) on fear recovery index (the difference between the first test trial and the last extinction trial for CS1+ and CS2+), which revealed a significant effect of reminder ($F_{1,71} = 8.920$, $P = 0.004$, $\eta^2 = 0.112$), cTBS ($F_{1,71} = 7.180$, $P = 0.009$, $\eta^2 = 0.092$) and reminder x cTBS interaction ($F_{1,71} = 10.613$, $P = 0.002$, $\eta^2 = 0.130$), but no effect of CS+ ($F_{1,71} = 0.518$, $P = 0.474$, $\eta^2 = 0.007$) or other interactions (all P s > 0.1;

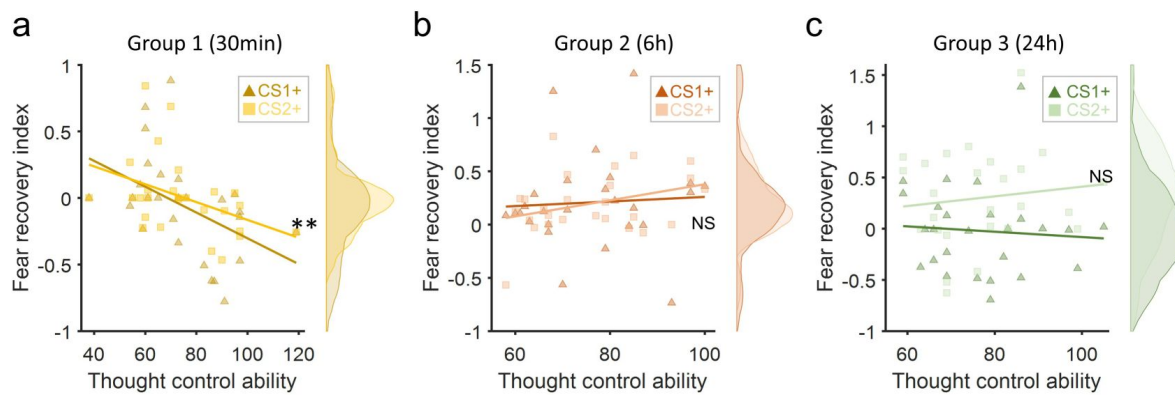


Fig. 3 |

Fear recovery as a function of thought-control abilities.

(a) Thought control ability was significantly correlated with fear recovery index in the 30min group ($P = 0.003$, Bonferroni correction), but not in the 6 hour (b) or 24 hour group (c, $P_s > 0.7$). The violin graphs indicate the distribution of fear recovery index across subjects (Fig. 2e). ** $P < 0.01$. NS: Non-significant.

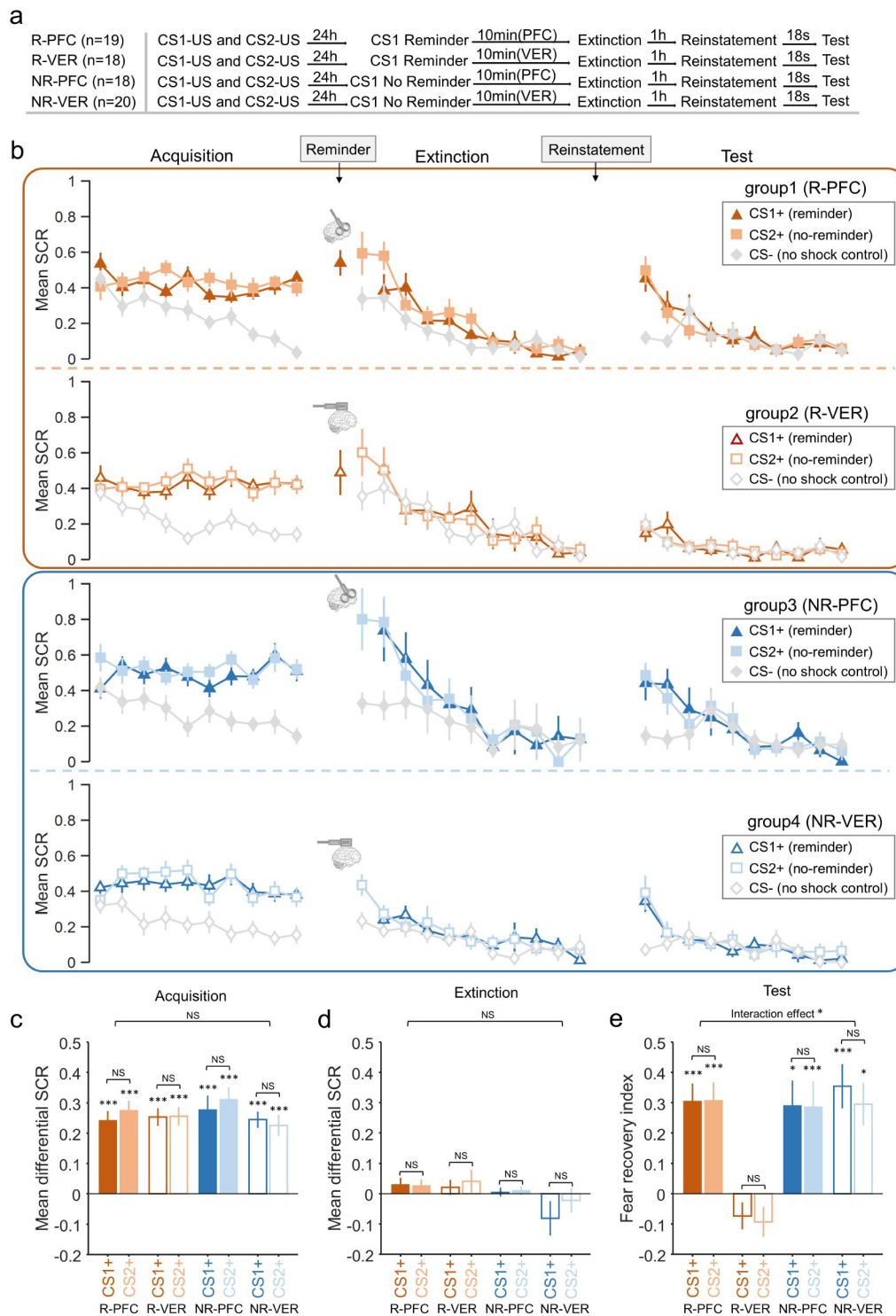


Fig. 4 |

SCR responses to conditioned stimuli in four cTBS groups.

a, Experimental design and timeline. **b**, Mean SCRs of fear conditioned stimuli CS1+ (reminder) and CS2+ (No-reminder), and the control stimulus (CS-) across the fear acquisition, extinction and test phases for each group (R - PFC, R-VER, NR-PFC and NR-VER groups). **c**, Mean differential SCRs (CS1+ minus CS- and CS2+ minus CS-) in the acquisition phase (latter half trials). **d**, Mean differential SCRs in the extinction phase (last trial). **e**, Differential fear recovery index between CS+ and CS- in the test phase. *** $P < 0.001$. * $P < 0.05$. NS: Non-significant. Error bars represent standard errors.

Fig. 4c). Post-hoc *t*-tests showed that there were significant fear reinstatement responses in the R-PFC group (CS1+: $t_{18} = 5.182$, $P < 0.001$; CS2+: $t_{18} = 5.258$, $P < 0.001$), the NR-PFC group (CS1+: $t_{17} = 3.496$, $P = 0.024$; CS2+: $t_{17} = 4.628$, $P < 0.001$) and the NR-VER group (CS1+: $t_{19} = 4.885$, $P < 0.001$; CS2+: $t_{19} = 3.262$, $P = 0.016$), but not in the R-VER group (CS1+: $t_{17} = -1.640$, $P = 0.119$; CS2+: $t_{17} = -0.744$, $P = 0.467$) (**Fig. 4e**), suggesting that both memory retrieval and normal dlPFC function are necessary for the short-term fear amnesia.

Finally, we went on to examine whether the fear reinstatement responses was related to the thought-control ability. A two-way ANOVA of reminder (reminder vs. no-reminder) $\times$ cTBS (rdlPFC vs. vertex) confirmed that there were no significant effects of reminder ($F_{1,71} = 0.365$, $P = 0.547$, $\eta^2 = 0.005$), cTBS ($F_{1,71} = 0.606$, $P = 0.439$, $\eta^2 = 0.008$) or their interaction ($F_{1,71} = 1.216$, $P = 0.274$, $\eta^2 = 0.017$) on the TCAQ scores. We then conducted simple linear regressions on different groups (as in study 2) and found a significant correlation between the thought-control ability and fear recovery index in the R-VER group (**Fig. 5b**, $t_{33} = -3.283$, $P = 0.008$, Bonferroni correction across 4 groups), consistent with the results of the short term (30min) amnesia effect in study 2 (**Fig. 3a**). Further post-hoc analysis of the individual CS+ showed that the correlations between thought-control ability and fear recovery index were significant for both CS+ (CS1+: $r = -0.557$, $P = 0.016$; CS2+: $r = -0.483$, $P = 0.042$). However, there was no such correlation in the R-PFC group (**Fig. 5a**, $t_{35} = 0.719$, $P = 0.477$), the NR-PFC group (**Fig. 5c**, $t_{33} = -1.334$, $P = 0.192$) or the NR-VER group (**Fig. 5d**, $t_{37} = -0.025$, $P = 0.980$) and the interaction of thought-control ability and CS+ effect was not significant in all 4 groups (P s > 0.3). Importantly, there was no significant correlation between thought-control ability and SCR in the late phase of acquisition or the last trial of extinction in four groups (all P s > 0.10), suggesting a specific relationship between thought-control ability and short-term fear recovery.

Discussion

Our results provide direct evidence to support the hypothesis that the memory reactivation might engage distinct cognitive mechanisms for fear memory modulation, which can be separated by their temporal dynamics, cue specificity and the dependence of thought-control ability. In line with previous research on the reconsolidation of fear memory³¹, our research replicated the classic results that behavioral extinction training during reconsolidation window specifically blocked the fear responses to the reminded CS+ while leaving non-reminded CS+ response intact (cue-specificity) in the long-term test (**Fig. 6a&b**). Moreover, thought-control ability was not related to the fear memory reconsolidation effect (**Fig. 6c**). However, findings of the short-term fear amnesia effect suggest that the reconsolidation framework may fall short to accommodate this more immediate effect (**Fig. 6a&b**).

Research in both declarative and associative emotional memories suggest that memory retrieval is not a simple act of obtaining a copy of stored information, but provides a critical window where the old memory is eligible to be integrated with new information¹⁴. Research in the field of declarative memory has shown that the motivated forgetting effect can appear early (10-30 minutes) after the memory suppression practices such as the influential think/no-think (TNT) paradigm¹⁵, suggesting that memory retrieval may indeed also facilitate short-term memory update^{32, 33}. Unlike the slow protein synthesis process involved in memory reconsolidation, the motivated forgetting effect was accompanied by heightened dorsolateral prefrontal cortex (dlPFC) activity, diminished hippocampal activation and altered functional connectivity between these brain regions and hence more temporally adaptive^{34, 35}.

Previous studies indicate that a suppression mechanism can be characterized by three distinct features: first, the memory suppression effect tends to emerge early, usually 10-30 mins after memory suppression practice and can be transient^{36, 37}; second, the memory suppression practice seems to directly act upon the unwanted memory itself³⁸, rendering it cue-independent

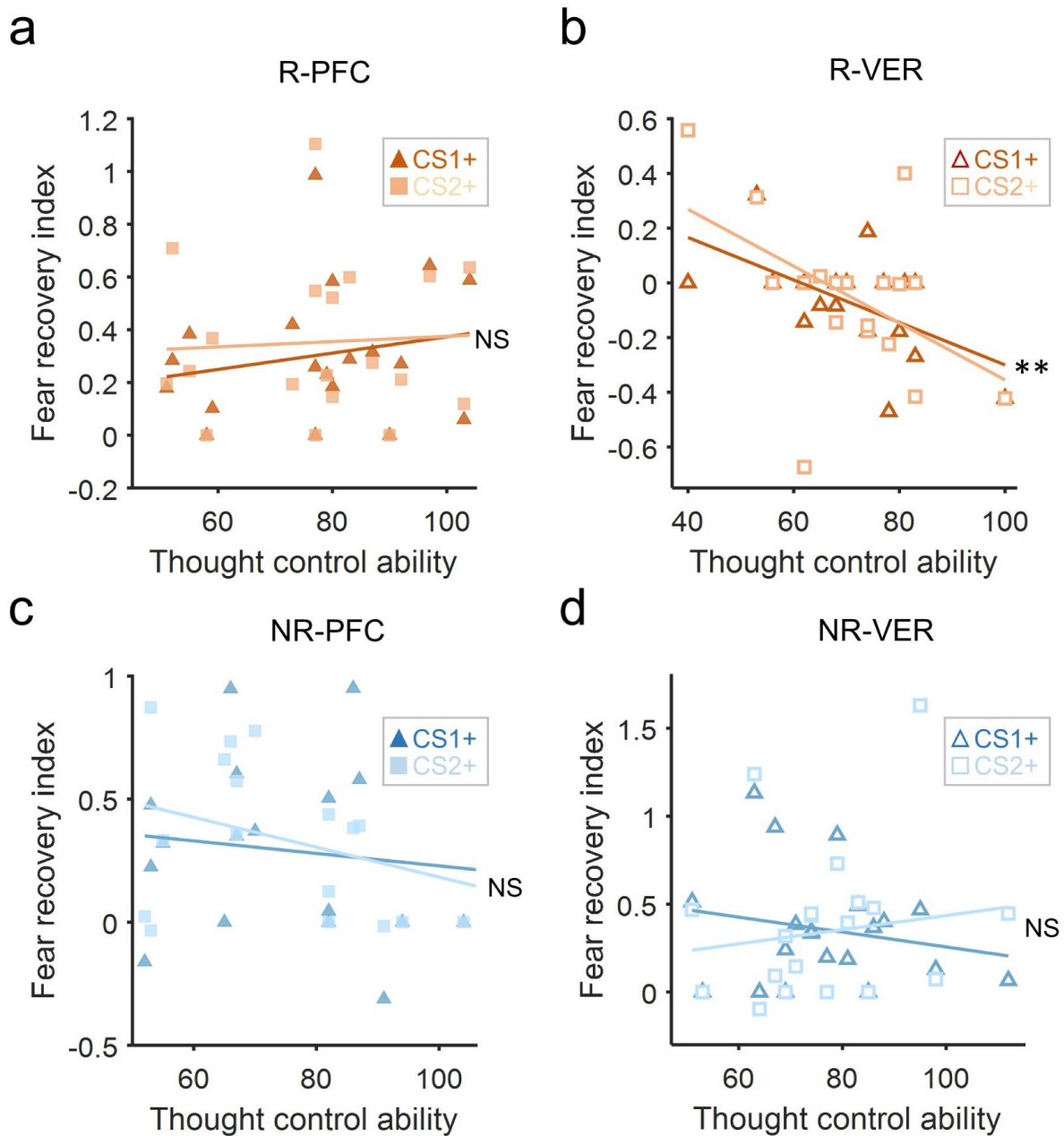


Fig. 5 |

The correlation between fear recovery index and thought-control abilities in four ctBS groups.

In the R-VER group, the correlation between thought control ability scores and fear recovery index was significant ($P = 0.008$, Bonferroni correction), with high thought-control ability participants showing less fear recovery for both CS+; whereas there was no significant correlation between fear recovery index and thought-control abilities in other three groups ($P_s > 0.4$, R-PFC, NR-PFC and NR-VER groups). ** $P < 0.01$, NS: Non-significant.

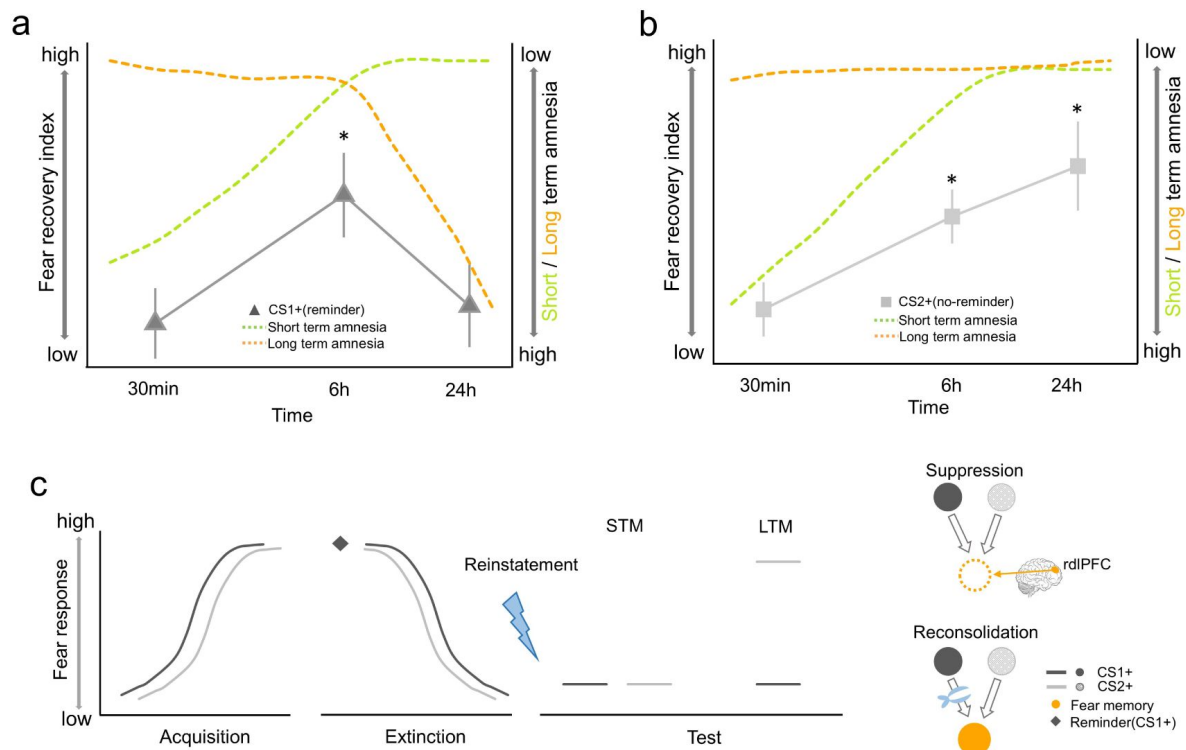


Fig. 6 |

Time courses of fear memory suppression and reconsolidation.

a, At the short interval (30min to 1h), fear recovery of the reminded CS (CS1+) is inhibited (green). As time progresses (from 6h to 24h), amnesia effect is mainly due to the later emerged fear memory reconsolidation effect (orange). The actual SCR data in black solid line. **b**, Fear memory suppression of the non-reminded CS (CS2+) is only evident at the 30-min interval and such effect starts to decay as test interval increases (green). However, fear reconsolidation does not affect CS2+ due to the cue-specificity of the reconsolidation effect (orange). The observed SCR data (black solid line) paralleled the prediction from both the short-term and the reconsolidation effects as the interval length increased (from 30min to 24h). **c**, Schematics demonstrating the effect of cue-reminder on fear memory retention. After both CS1+ (black) and CS2+ (grey) successfully elicit fear responses in the acquisition phase, CS1+ is reminded (black) before both CS+ go through the extinction training. The lack of both CS1+ and CS2+ fear responses in the short-term memory (STM) test (30 min) can be explained by the dlPFC dependent direct suppression effect (dotted circle of US representation). However, the cue-specific fear amnesia effect in the long-term memory (LTM) test (24h) of CS1+ but not CS2+ is attributed to the reconsolidation effect specific to CS1+. * $P < 0.05$, Error bars represent standard errors.

such that other cues originally associated with the unwanted memory also fail in memory recall; third, the magnitude of memory suppression effects is believed to be associated with individual difference in control abilities over intrusive thoughts¹⁷. The short-term fear dementia effects we identified are fundamentally different from the memory reconsolidation effects and fit closely with the characteristics of the memory suppression effect. Inspired by the similarities between our results and suppression-induced declarative memory amnesia³⁹, we hypothesize that the retrieval-extinction procedure might facilitate a spontaneous memory suppression process and thus yield a short-term amnesia effect. Accordingly, the activated fear memory induced by the retrieval cue would be subjected to an automatic fear memory suppression through the extinction training¹⁶.

In our experiments, subjects were not explicitly instructed to suppress their fear expression, yet the retrieval-extinction training significantly decreased short-term fear expression. These results are consistent with the short-term dementia induced with the more explicit suppression intervention^{11–13, 15, 35}. It is worth noting that the standard memory suppression paradigm involve consciously repelling unwanted memory, which might not be necessary to engage the suppression mechanism. For example, in the well-known retrieval-induced forgetting (RIF) phenomenon, the recall of a stored memory can impair the retention of related long-term memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner⁴⁰. Moreover, subjects with greater history of trauma exhibited more suppression-induced forgetting of both negative and neutral memories than those with little or no trauma⁴¹. Similarly, people with higher self-reported thought-control capabilities showed more severe cue-independent recall deficit, suggesting that suppression mechanism is associated with individual differences in spontaneous control abilities over intrusive thoughts¹⁷. It has been suggested that similar automatic mechanisms might be involved in organic retrograde amnesia of the traumatic childhood memories^{42, 43}.

Taken together, we propose that similar brain mechanisms, such as the dynamic brain activity and connectivity modulation, involved in memory suppression may be deployed spontaneously and act as a general cognitive mechanism to cope with intrusive memories in daily life. Indeed, recently researchers have proposed that the standard retrieval-extinction paradigm to study fear memory might also engage automatic memory suppression¹⁶. Consistent with this hypothesis, we showed that the retrieval cue was necessary to produce the short-term amnesia effect (**Figs. 1** & **2**) and critically, our results fitted well with the three key characteristics of the active suppression mechanism: temporal dynamics, cue-specificity and the dependence of thought-control ability (**Figs. 2** & **3**).

To calibrate the temporal boundaries of the short and long-term effects, we also included the medium-term group where fear expression was tested 6 hours after the retrieval-extinction training. Surprisingly, our results showed that fear memory returned at this time point (**Fig. 6a&b**), consistent with previous animal and human studies that suggest the 6-hour lower time boundary for memory reconsolidation^{4, 8–10, 44}. However, the return of fear expression in the medium-term group also indicates that the short-term fear amnesia is transient, laying out the upper time boundary for the hypothesized fear memory suppression effect. Although mixed results have been reported regarding the durability of suppression effects in the declarative memory studies^{45, 46}, future research will be needed to investigate whether the short-term effect we observed is specifically related to associative memory or the spontaneous nature of suppression. Nevertheless, our results provide clear evidence that the short-term amnesia effect can be temporally separated from the reconsolidation effect (**Fig. 6c**). Previous research also showed that separate mnemonic processes can be involved after emotional memory retrieval. For example, by varying the duration length and repetition number of memory cue re-exposure, different levels of memory persistence (reconsolidation, extinction or a “limbo” state in between) can be induced, suggesting a neural titration mechanism for memory de-stabilization and re-

stabilization ⁴⁷, ⁴⁸. These works identified important “boundary conditions” of memory retrieval in affecting the retention of the maladaptive emotional memories. In our study, however, we showed that even within a boundary condition previously thought to elicit memory reconsolidation, mnemonic processes other than reconsolidation could also be at work, and these processes jointly shape the persistence of fear memory.

Interestingly, our results also corroborate with a recent study showing that blocking fear reconsolidation using post-retrieval repetitive transcranial magnetic stimulation (rTMS) did not influence the short-term fear memory expression ⁴⁹. However, as we demonstrated in studies 2 & 3, memory retrieval and intact dlPFC function were both necessary for the short-term reinstatement fear amnesia after the extinction training. Therefore, instead of “intact short-term fear expression by the rTMS over the dlPFC”, their results could also be interpreted such that the rTMS on the dlPFC demolished the otherwise short-term fear amnesia induced by the memory retrieval cue. Indeed, a series of neuroimaging studies have revealed that the prefrontal neural network including anterior cingulate cortex, anterior ventrolateral prefrontal cortex and the dlPFC is directly engaged in purging the unwanted declarative memory from consciousness ³³, ³⁴, ³⁸. Moreover, previous research has established a causal relationship between dlPFC activity and the memory suppression mechanism underlying retrieval-induced forgetting using transcranial direct current stimulation (tDCS) ^{50–52}. In our experiment (studies 2&3), when the dlPFC function was intact, subjects’ thought-control abilities were positively correlated with fear amnesia in the test phase. Such correlations disappeared in the no-reminder groups (NR-VER and NR-PFC) and the R-PFC group (**Figs. 3** & **5**), establishing a causal link between the dlPFC activity and short-term fear amnesia.

These results help shed light on the dynamics of memory modulation after memory activation and designing novel treatment of psychiatric disorders caused by excessive fear or anxiety ⁵³. Memory reconsolidation and suppression both have been studied thoroughly in separate fields. Specifically, aversive memories are usually associated with different retrieval cues, and interfering memory reconsolidation only blocks learned fear memory associated with the reactivated memory probe but not the others ⁴, ⁸–¹⁰, ⁴⁴. On the other hand, spontaneous memory suppression is cue-independent and directly suppresses the fear memory trace yet only effective within limited durability ³⁶. Although future research is clearly needed to understand the brain mechanisms of the short-term effect and its connection to the memory suppression effect, especially in relevance to clinical populations such as posttraumatic stress disorder (PTSD) and phobia patients ⁵⁴, ⁵⁵, our results provide a general framework to highlight the potential of memory retrieval to triggering different memory updating mechanisms with unique temporal dynamics.

Transparency

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The authors have declared no conflict of interests.

Data availability

All the data of this study are available for download at <https://osf.io/9agvk/>.

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Editors

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Reviewer #1 (Public Review):

Summary:

The novel advance by Wang et al is in the demonstration that, relative to a standard extinction procedure, the retrieval-extinction procedure more effectively suppresses responses to a conditioned threat stimulus when testing occurs just minutes after extinction. The authors provide some solid evidence to show that this "short-term" suppression of responding involves engagement of the dorsolateral prefrontal cortex.

Strengths:

Overall, the study is well-designed and the results are potentially interesting. There are, however, a few issues in the way that it is introduced and discussed. Some of the issues concern clarity of expression/communication. However, others relate to a theory that could be used to help the reader understand why the results should have come out the way that they did. More specific comments and questions are presented below.

Weaknesses:

INTRODUCTION & THEORY

(1) Can the authors please clarify why the first trial of extinction in a standard protocol does NOT produce the retrieval-extinction effect? Particularly as the results section states: "Importantly, such a short-term effect is also retrieval dependent, suggesting the labile state of memory is necessary for the short-term memory update to take effect (Fig. 1e)." The importance of this point comes through at several places in the paper:

1A. "In the current study, fear recovery was tested 30 minutes after extinction training, whereas the effect of memory reconsolidation was generally evident only several hours later and possibly with the help of sleep, leaving open the possibility of a different cognitive mechanism for the short-term fear dementia related to the retrieval-extinction procedure."

***What does this mean? The two groups in study 1 experienced a different interval between the first and second CS extinction trials; and the results varied with this interval: a longer interval (10 min) ultimately resulted in less reinstatement of fear than a shorter interval. Even if the different pattern of results in these two groups was shown/known to imply two different processes, there is absolutely no reason to reference any sort of cognitive mechanism or dementia - that is quite far removed from the details of the present study.

1B. "Importantly, such a short-term effect is also retrieval dependent, suggesting the labile state of memory is necessary for the short-term memory update to take effect (Fig. 1e)." ***As above, what is "the short-term memory update"? At this point in the text, it would be appropriate for the authors to discuss why the retrieval-extinction procedure produces less recovery than a standard extinction procedure as the two protocols only differ in the interval between the first and second extinction trials. References to a "short-term memory update" process do not help the reader to understand what is happening in the protocol.

(2) "Indeed, through a series of experiments, we identified a short-term fear amnesia effect following memory retrieval, in addition to the fear reconsolidation effect that appeared much later."

***The only reason for supposing two effects is because of the differences in responding to the CS2, which was subjected to STANDARD extinction, in the short- and long-term tests. More needs to be said about how and why the performance of CS2 is affected in the short-term test and recovers in the long-term test. That is, if the loss of performance to CS1 and CS2 is going to be attributed to some type of memory updating process across the retrieval-extinction procedure, one needs to explain the selective recovery of performance to CS2 when the extinction-to-testing interval extends to 24 hours. Instead of explaining this recovery, the authors note that performance to CS1 remains low when the extinction-to-testing interval is 24 hours and invoke something to do with memory reconsolidation as an explanation for their results: that is, they imply (I think) that reconsolidation of the CS1-US memory is disrupted across the 24-hour interval between extinction and testing even though CS1 evokes negligible responding just minutes after extinction.

(3) The discussion of memory suppression is potentially interesting but, in its present form, raises more questions than it answers. That is, memory suppression is invoked to explain a particular pattern of results but I, as the reader, have no sense of why a fear memory would be better suppressed shortly after the retrieval-extinction protocol compared to the standard

extinction protocol; and why this suppression is NOT specific to the cue that had been subjected to the retrieval-extinction protocol.

3A. Relatedly, how does the retrieval-induced forgetting (which is referred to at various points throughout the paper) relate to the retrieval-extinction effect? The appeal to retrieval-induced forgetting as an apparent justification for aspects of the present study reinforces points 2 and 3 above. It is not uninteresting but needs some clarification/elaboration.

(4) Given the reports by Chalkia, van Oudenhove & Beckers (2020) and Chalkia et al (2020), some qualification needs to be inserted in relation to reference 6. That is, reference 6 is used to support the statement that "during the reconsolidation window, old fear memory can be updated via extinction training following fear memory retrieval". This needs a qualifying statement like "[but see Chalkia et al (2020a and 2020b) for failures to reproduce the results of 6]."

<https://pubmed.ncbi.nlm.nih.gov/32580869/>
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7115860/>

CLARIFICATIONS, ELABORATIONS, EDITS

(5) The Abstract was not easy to follow:

5A. What does it mean to ask: "whether memory retrieval facilitates update mechanisms other than memory reconsolidation"? That is, in what sense could or would memory retrieval be thought to facilitate a memory update mechanism?

5B. "First, we demonstrate that memory reactivation prevents the return of fear shortly after extinction training in contrast to the memory reconsolidation effect which takes several hours to emerge and such a short-term amnesia effect is cue independent (Study 1, N = 57 adults)."

***The phrasing here could be improved for clarity: "First, we demonstrate that the retrieval-extinction protocol prevents the return of fear shortly after extinction training (i.e., when testing occurs just min after the end of extinction)." Also, cue-dependence of the retrieval-extinction effect was assessed in study 2.

5C. "Furthermore, memory reactivation also triggers fear memory reconsolidation and produces cue-specific amnesia at a longer and separable timescale (Study 2, N = 79 adults)."

***In study 2, the retrieval-extinction protocol produced a cue-specific disruption in responding when testing occurred 24 hours after the end of extinction. This result is interesting but cannot be easily inferred from the statement that begins "Furthermore..." That is, the results should be described in terms of the combined effects of retrieval and extinction, not in terms of memory reactivation alone; and the statement about memory reconsolidation is unnecessary. One can simply state that the retrieval-extinction protocol produced a cue-specific disruption in responding when testing occurred 24 hours after the end of extinction.

5D. "...we directly manipulated brain activities in the dorsolateral prefrontal cortex and found that both memory retrieval and intact prefrontal cortex functions were necessary for the short-term fear amnesia."

***This could be edited to better describe what was shown: E.g., "...we directly manipulated brain activities in the dorsolateral prefrontal cortex and found that intact prefrontal cortex functions were necessary for the short-term fear amnesia after the retrieval-extinction protocol."

5E. "The temporal scale and cue-specificity results of the short-term fear amnesia are clearly dissociable from the amnesia related to memory reconsolidation, and suggest that memory retrieval and extinction training trigger distinct underlying memory update mechanisms."

***The pattern of results when testing occurred just minutes after the retrieval-extinction

protocol was different from that obtained when testing occurred 24 hours after the protocol. Describing this in terms of temporal scale is unnecessary, and suggesting that memory retrieval and extinction trigger different memory update mechanisms is not obviously warranted. The results of interest are due to the combined effects of retrieval+extinction and there is no sense in which different memory update mechanisms should be identified with retrieval (mechanism 1) and extinction (mechanism 2).

5F. "These findings raise the possibility of concerted memory modulation processes related to memory retrieval..."

***What does this mean?

(6) "...suggesting that the fear memory might be amenable to a more immediate effect, in addition to what the memory reconsolidation theory prescribes..."

***What does it mean to say that the fear memory might be amenable to a more immediate effect?

(7) "Parallel to the behavioral manifestation of long- and short-term memory deficits, concurrent neural evidence supporting memory reconsolidation theory emphasizes the long-term effect of memory retrieval by hypothesizing that synapse degradation and de novo protein synthesis are required for reconsolidation."

***This sentence needs to be edited for clarity.

(8) "previous behavioral manipulations engendering the short-term declarative memory effect..."

***What is the declarative memory effect? It should be defined.

(9) "The declarative amnesia effect emerges much earlier due to the online functional activity modulation..."

***Even if the declarative memory amnesia effect had been defined, the reference to online functional activity modulation is not clear.

(10) "However, it remains unclear whether memory retrieval might also precipitate a short-term amnesia effect for the fear memory, in addition to the long-term prevention orchestrated by memory consolidation."

***I found this sentence difficult to understand on my first pass through the paper. I think it is because of the phrasing of memory retrieval. That is, memory retrieval does NOT precipitate any type of short-term amnesia for the fear memory: it is the retrieval-extinction protocol that produces something like short-term amnesia. Perhaps this sentence should also be edited for clarity.

I will also note that the usage of "short-term" at this point in the paper is quite confusing: Does the retrieval-extinction protocol produce a short-term amnesia effect, which would be evidenced by some recovery of responding to the CS when tested after a sufficiently long delay? I don't believe that this is the intended meaning of "short-term" as used throughout the majority of the paper, right?

(11) "To fully comprehend the temporal dynamics of the memory retrieval effect..."

***What memory retrieval effect? This needs some elaboration.

(12) "We hypothesize that the labile state triggered by the memory retrieval may facilitate different memory update mechanisms following extinction training, and these mechanisms can be further disentangled through the lens of temporal dynamics and cue-specificities."

***What does this mean? The first part of the sentence is confusing around the usage of the term "facilitate"; and the second part of the sentence that references a "lens of temporal dynamics and cue-specificities" is mysterious. Indeed, as all rats received the same retrieval-extinction exposures in Study 2, it is not clear how or why any differences between the groups are attributed to "different memory update mechanisms following extinction".

(13) "In the first study, we aimed to test whether there is a short-term amnesia effect of fear memory retrieval following the fear retrieval-extinction paradigm."

***Again, the language is confusing. The phrase, "a short-term amnesia effect" implies that the amnesia itself is temporary; but I don't think that this implication is intended. The problem is specifically in the use of the phrase "a short-term amnesia effect of fear memory retrieval." To the extent that short-term amnesia is evident in the data, it is not due to retrieval per se but, rather, the retrieval-extinction protocol.

(14) The authors repeatedly describe the case where there was a 24-hour interval between extinction and testing as consistent with previous research on fear memory reconsolidation. Which research exactly? That is, in studies where a CS re-exposure was combined with a drug injection, responding to the CS was disrupted in a final test of retrieval from long-term memory which typically occurred 24 hours after the treatment. Is that what the authors are referring to as consistent? If so, which aspect of the results are consistent with those previous findings? Perhaps the authors mean to say that, in the case where there was a 24-hour interval between extinction and testing, the results obtained here are consistent with previous research that has used the retrieval-extinction protocol. This would clarify the intended meaning greatly.

DATA

(15) Points about data:

15A. The eight participants who were discontinued after Day 1 in study 1 were all from the no-reminder group. Can the authors please comment on how participants were allocated to the two groups in this experiment so that the reader can better understand why the distribution of non-responders was non-random (as it appears to be)?

15B. Similarly, in study 2, of the 37 participants that were discontinued after Day 2, 19 were from Group 30 min, and 5 were from Group 6 hours. Can the authors comment on how likely these numbers are to have been by chance alone? I presume that they reflect something about the way that participants were allocated to groups, but I could be wrong.

15C. "Post hoc t-tests showed that fear memories were resilient after regular extinction training, as demonstrated by the significant difference between fear recovery indexes of the CS+ and CS- for the no-reminder group ($t_{26} = 7.441$, $P < 0.001$; Fig. 1e), while subjects in the reminder group showed no difference of fear recovery between CS+ and CS- ($t_{29} = 0.797$, $P = 0.432$, Fig. 1e)."

***Is the fear recovery index shown in Figure 1E based on the results of the 1st test trial only? How can there have been a "significant difference between fear recovery indexes of the CS+ and CS- for the no-reminder group" when the difference in responding to the CS+ and CS- is used to calculate the fear recovery index shown in 1E? What are the t-tests comparing exactly, and what correction is used to account for the fact that they are applied post-hoc?

15D. "Finally, there is no statistical difference between the differential fear recovery indexes between CS+ in the reminder and no reminder groups ($t_{55} = -2.022$, $P = 0.048$; Fig. 1c, also see Supplemental Material for direct test for the test phase)."

***Is this statement correct - i.e., that there is no statistically significant difference in fear recovery to the CS+ in the reminder and no reminder groups? I'm sure that the authors would like to claim that there IS such a difference; but if such a difference is claimed, one would be concerned by the fact that it is coming through in an uncorrected t-test, which is the third one of its kind in this paragraph. What correction (for the Type 1 error rate) is used to account for the fact that the t-tests are applied post-hoc? And if no correction, why not?

15E. In study 2, why is responding to the CS- so high on the first test trial in Group 30 min? Is the change in responding to the CS- from the last extinction trial to the first test trial different

across the three groups in this study? Inspection of the figure suggests that it is higher in Group 30 min relative to Groups 6 hours and 24 hours. If this is confirmed by the analysis, it has implications for the fear recovery index which is partly based on responses to the CS-. If not for differences in the CS- responses, Groups 30 minutes and 6 hours are otherwise identical.

15F. Was the 6-hour group tested at a different time of day compared to the 30-minute and 24-hour groups; and could this have influenced the SCRs in this group?

15G. Why is the range of scores in "thought control ability" different in the 30-minute group compared to the 6-hour and 24-hour groups? I am not just asking about the scale on the x-axis: I am asking why the actual distribution of the scores in thought control ability is wider for the 30-minute group?

(16) During testing in each experiment, how were the various stimuli presented? That is, was the presentation order for the CS+ and CS- pseudorandom according to some constraint, as it had been in extinction? This information should be added to the method section.

(17) "These results are consistent with previous research which suggested that people with better capability to resist intrusive thoughts also performed better in motivated dementia in both declarative and associative memories."

***Which parts of the present results are consistent with such prior results? It is not clear from the descriptions provided here why thought control ability should be related to the present findings or, indeed, past ones in other domains. This should be elaborated to make the connections clear.

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Reviewer #2 (Public Review):

Summary

The study investigated whether memory retrieval followed soon by extinction training results in a short-term memory deficit when tested - with a reinstatement test that results in recovery from extinction - soon after extinction training. Experiment 1 documents this phenomenon using a between-subjects design. Experiment 2 used a within-subject control and saw that the effect was also observed in a control condition. In addition, it also revealed that if testing is conducted 6 hours after extinction, there is no effect of retrieval prior to extinction as there is recovery from extinction independently of retrieval prior to extinction. A third group also revealed that retrieval followed by extinction attenuates reinstatement when the test is conducted 24 hours later, consistent with previous literature. Finally, Experiment 3 used continuous theta-burst stimulation of the dorsolateral prefrontal cortex and assessed whether inhibition of that region (vs a control region) reversed the short-term effect revealed in Experiments 1 and 2. The results of the control groups in Experiment 3 replicated the previous findings (short-term effect), and the experimental group revealed that these can be reversed by inhibition of the dorsolateral prefrontal cortex.

Strengths

The work is performed using standard procedures (fear conditioning and continuous theta-burst stimulation) and there is some justification for the sample sizes. The results replicate previous findings - some of which have been difficult to replicate and this needs to be acknowledged - and suggest that the effect can also be observed in a short-term reinstatement test.

The study establishes links between memory reconsolidation and retrieval-induced forgetting (or memory suppression) literature. The explanations that have been developed for these are distinct and the current results integrate these, by revealing that the DLPFC activity involved in retrieval-extinction short-term effect. There is thus some novelty in the present results, but numerous questions remain unaddressed.

Weakness

The fear acquisition data is converted to a differential fear SCR and this is what is analysed (early vs late). However, the figure shows the raw SCR values for CS+ and CS- and therefore it is unclear whether the acquisition was successful (despite there being an "early" vs "late" effect - no descriptives are provided).

In Experiment 1 (Test results) it is unclear whether the main conclusion stems from a comparison of the test data relative to the last extinction trial ("we defined the fear recovery index as the SCR difference between the first test trial and the last extinction trial for a specific CS") or the difference relative to the CS- ("differential fear recovery index between CS+ and CS-"). It would help the reader assess the data if Figure 1e presents all the indexes (both CS+ and CS-). In addition, there is one sentence that I could not understand "there is no statistical difference between the differential fear recovery indexes between CS+ in the reminder and no reminder groups ($P=0.048$)". The p-value suggests that there is a difference, yet it is not clear what is being compared here. Critically, any index taken as a difference relative to the CS- can indicate recovery of fear to the CS+ or absence of discrimination relative to the CS-, so ideally the authors would want to directly compare responses to the CS+ in the reminder and no-reminder groups. The latter issue is particularly relevant in Experiment 2, in which the CS- seems to vary between groups during the test and this can obscure the interpretation of the result.

In Experiment 1, the findings suggest that there is a benefit of retrieval followed by extinction in a short-term reinstatement test. In Experiment 2, the same effect is observed on a cue that did not undergo retrieval before extinction (CS2+), a result that is interpreted as resulting from cue-independence, rather than a failure to replicate in a within-subjects design the observations of Experiment 1 (between-subjects). Although retrieval-induced forgetting is cue-independent (the effect on items that are suppressed [Rp-] can be observed with an independent probe), it is not clear that the current findings are similar. Here, both cues have been extinguished and therefore been equally exposed during the critical stage.

The findings in Experiment 2 suggest that the amnesia reported in Experiment 1 is transient, in that no effect is observed when the test is delayed by 6 hours. The phenomena whereby reactivated memories transition to extinguished memories as a function of the amount of exposure (or number of trials) is completely different from the phenomena observed here. In the former, the manipulation has to do with the number of trials (or the total amount of time) that the cues are exposed to. In the current study, the authors did not manipulate the number of trials but instead the retention interval between extinction and test. The finding reported here is closer to a "Kamin effect", that is the forgetting of learned information which is observed with intervals of intermediate length (Baum, 1968). Because the Kamin effect has been inferred to result from retrieval failure, it is unclear how this can be explained here. There needs to be much more clarity on the explanations to substantiate the conclusions.

There are many results (Ryan et al., 2015) that challenge the framework that the authors base their predictions on (consolidation and reconsolidation theory), therefore these need to be acknowledged. Similarly, there are reports that failed to observe the retrieval-extinction phenomenon (Chalkia et al., 2020), and the work presented here is written as if the phenomenon under consideration is robust and replicable. This needs to be acknowledged.

The parallels between the current findings and the memory suppression literature are speculated in the general discussion, and there is the conclusion that "the retrieval-extinction procedure might facilitate a spontaneous memory suppression process". Because one of the basic tenets of the memory suppression literature is that it reflects an "active suppression" process, there is no reason to believe that in the current paradigm, the same phenomenon is in place, but instead, it is "automatic". In other words, the conclusions make strong parallels with the memory suppression (and cognitive control) literature, yet the phenomena that they observed are thought to be passive (or spontaneous/automatic).

Ultimately, it is unclear why 10 mins between the reminder and extinction learning will "automatically" suppress fear memories. Further down in the discussion, it is argued that "For example, in the well-known retrieval-induced forgetting (RIF) phenomenon, the recall of a stored memory can impair the retention of related long-term memory and this forgetting effect emerges as early as 20 minutes after the retrieval procedure, suggesting memory suppression or inhibition can occur in a more spontaneous and automatic manner". I did not follow with the time delay between manipulation and test (20 mins) would speak about whether the process is controlled or automatic.

Among the many conclusions, one is that the current study uncovers the "mechanism" underlying the short-term effects of retrieval extinction. There is little in the current report that uncovers the mechanism, even in the most psychological sense of the mechanism, so this needs to be clarified. The same applies to the use of "adaptive".

Whilst I could access the data on the OFS site, I could not make sense of the Matlab files as there is no signposting indicating what data is being shown in the files. Thus, as it stands, there is no way of independently replicating the analyses reported.

The supplemental material shows figures with all participants, but only some statistical analyses are provided, and sometimes these are different from those reported in the main manuscript. For example, the test data in Experiment 1 is analysed with a two-way ANOVA with the main effects of group (reminder vs no-reminder) and time (last trial of extinction vs first trial of the test) in the main report. The analyses with all participants in the sup mat used a mixed two-way ANOVA with a group (reminder vs no reminder) and CS (CS+ vs CS-). This makes it difficult to assess the robustness of the results when including all participants. In addition, in the supplementary materials, there are no figures and analyses for Experiment 3.

One of the overarching conclusions is that the "mechanisms" underlying reconsolidation (long term) and memory suppression (short term) phenomena are distinct, but memory suppression phenomena can also be observed after a 7-day retention interval (Storm et al., 2012), which then questions the conclusions achieved by the current study.

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Reviewer #3 (Public Review):

SUMMARY

Wang et al. have addressed how acquired fear and extinction memories evolve over time. Using a retrieval-extinction procedure in healthy humans, they have investigated the recovery of fear memories 30-60 minutes, 6 hours, and 24 hours after the retrieval-extinction phase. They have addressed this research question through 3 different experiments which included manipulations of the reminder cue, the time interval, and brain activity. Together, the studies suggest that early on after retrieval-extinction (30-60 min. later), retrieval-extinction may lead to an attenuation of fear recovery (after reinstatement) for all fear cues, as well as the non-reminded ones. Study 3 moreover suggests that this effect may depend on normal dlPFC function. In addition, the paper also contains data in line with prior findings suggesting that a 6-hour interval does not benefit from the reminder cue, and that a 24-hour interval does, and specifically for the reminded fear cue. The latter findings are seen as evidence of fear memory reconsolidation.

STRENGTHS

- (1) The paper combines three related human fear conditioning studies, each with decent sample sizes. The authors are transparent about the fact that they excluded many participants and about which conditions they belonged to.
- (2) The effect that this paper investigates (short-term fear memory after a retrieval-extinction procedure) has not been studied extensively, thus making it a relevant topic.
- (3) The application of brain stimulation as a means to study causal relationships is interesting and goes beyond the purely behavioral or pharmacological interventions that are often used in human fear conditioning research. Also, the use of an active control stimulation is a strength of the study.

WEAKNESSES

- (1) The entire study hinges on the idea that there is memory 'suppression' if (1) the CS+ was reminded before extinction and (2) the reinstatement and memory test takes place 30 minutes later (in Studies 1 & 2). However, the evidence supporting this suppression idea is not very strong. In brief, in Study 1, the effect seems to only just reach significance, with a medium effect size at best, and, moreover, it is unclear if this is the correct analysis (which is a bit doubtful, when looking at Figure 1D and E). In Study 2, there was no optimal control condition without reminder and with the same 30-min interval (which is problematic, because we can assume generalization between CS1+ and CS2+, as pointed out by the authors, and because generalization effects are known to be time-dependent). Study 3 is more convincing, but entails additional changes in comparison with Studies 1 and 2, i.e., applications of cTBS and an interval of 1 hour instead of 30 minutes (the reason for this change was not explained). So, although the findings of the 3 studies do not contradict each other and are coherent, they do not all provide strong evidence for the effect of interest on their own.

Related to the comment above, I encourage the authors to double-check if this statement is correct: "Also, our results remain robust even with the "non-learners" included in the analysis (Fig. S1 in the Supplemental Material)". The critical analysis for Study 1 is a between-group comparison of the CS+ and CS- during the last extinction trial versus the first test trial. This result only just reached significance with the selected sample ($p = .048$), and Figures 1D and E even seem to suggest otherwise. I doubt that the analysis would reach significance when including the "non-learners" - assuming that this is what is shown in Supplemental Figure 1 (which shows the data from "all responded participants").

Also related to the comment above, I think that the statement "suggesting a cue-independent short-term amnesia effect" in Study 2 is not correct and should read: "suggesting extinction of fear to the CS1+ and CS2+", given that the response to the CS+'s is similar to the response to the CS-, as was the case at the end of extinction. Also the next statement "This result indicates that the short-term amnesia effect observed in Study 2 is not reminder-cue specific and can generalize to the non-reminded cues" is not fully supported by the data, given the lack of an appropriate control group in this study (a group without reinstatement). The comparison with the effect found in Study 1 is difficult because the effect found there was relatively small (and may have to be double-checked, see remarks above), and it was obtained with a different procedure using a single CS+. The comparison with the 6-h and 24-h groups of Study 2 is not helpful as a control condition for this specific question (i.e., is there reinstatement of fear for any of the CS+'s) because of the large procedural difference with regard to the intervals between extinction and reinstatement (test).

(2) It is unclear which analysis is presented in Figure 3. According to the main text, it either shows the "differential fear recovery index between CS+ and CS-" or "the fear recovery index of both CS1+ and CS2+". The authors should clarify what they are analyzing and showing, and clarify to which analyses the ** and NS refer in the graphs. I would also prefer the X-axes and particularly the Y-axes of Fig. 3a-b-c to be the same. The image is a bit misleading now. The same remarks apply to Figure 5.

(3) In general, I think the paper would benefit from being more careful and nuanced in how the literature and findings are represented. First of all, the authors may be more careful when using the term 'reconsolidation'. In the current version, it is put forward as an established and clearly delineated concept, but that is not the case. It would be useful if the authors could change the text in order to make it clear that the reconsolidation framework is a theory, rather than something that is set in stone (see e.g., Elsey et al., 2018 (<https://doi.org/10.1037/bul0000152>), Schroyens et al., 2022 (<https://doi.org/10.3758/s13423-022-02173-2>)).

In addition, the authors may want to reconsider if they want to cite Schiller et al., 2010 (<https://doi.org/10.1038/nature08637>), given that the main findings of this paper, nor the analyses could be replicated (see, Chalkia et al., 2020 (<https://doi.org/10.1016/j.cortex.2020.04.017>; <https://doi.org/10.1016/j.cortex.2020.03.031>)).

Relatedly, it should be clarified that Figure 6 is largely speculative, rather than a proven model as it is currently presented. This is true for all panels, but particularly for panel c, given that the current study does not provide any evidence regarding the proposed reconsolidation mechanism.

Lastly, throughout the paper, the authors equate skin conductance responses (SCR) with fear memory. It should at least be acknowledged that SCR is just one aspect of a fear response, and that it is unclear whether any of this would translate to verbal or behavioral effects. Such effects would be particularly important for any clinical application, which the authors put forward as the ultimate goal of the research.

(4) The Discussion quite narrowly focuses on a specific 'mechanism' that the authors have in mind. Although it is good that the Discussion is to the point, it may be worthwhile to entertain other options or (partial) explanations for the findings. For example, have the authors considered that there may be an important role for attention? When testing very soon after the extinction procedure (and thus after the reminder), attentional processes may play an important role (more so than with longer intervals). The retrieval procedure could perhaps induce heightened attention to the reminded CS+ (which could be further enhanced by dlPFC stimulation)?

(5) There is room for improvement in terms of language, clarity of the writing, and (presentation of the) statistical analyses, for all of which I have provided detailed feedback in

the 'Recommendations for the authors' section. Idem for the data availability; they are currently not publicly available, in contrast with what is stated in the paper. In addition, it would be helpful if the authors would provide additional explanation or justification for some of the methodological choices (e.g., the 18-s interval and why stimulate 8 minutes after the reminder cue, the choice of stimulation parameters), and comment on reasons for (and implications of) the large amount of excluded participants (>25%).

Finally, I think several statements made in the paper are overly strong in light of the existing literature (or the evidence obtained here) or imply causal relationships that were not directly tested.

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